

Activity 8: Predictive-maintenance Decision Design

Level: Advanced Duration: 75 minutes Mapping: K3 A5 Tools: Sensor CSV; FMEA worksheet; maintenance checklist

Goal

Combine asset criticality, condition data, work history and model evidence into a controlled maintenance decision.

Learner output

Criticality/FMEA screen, condition trend, action threshold, work-order recommendation and fallback.

Files in this folder

- README.md / README.pdf - activity guide and detailed procedure
- scenario.md / scenario.pdf - case facts and questions
- worksheet.md / worksheet.pdf - structured response template
- evidence-checklist.md / evidence-checklist.pdf - acceptance record
- data.csv - mock dataset when analysis is required

Detailed procedure

1. Confirm equipment function, safety consequence and production criticality.
2. Review sensor-history.csv with maintenance and operating context.
3. Plot or inspect vibration, temperature and risk-score trends.
4. List plausible failure modes and current preventive/detection controls.
5. Define normal, alert and stop bands using engineering evidence, not the model score alone.
6. Draft a work-order recommendation with source references and required inspection.
7. Define who approves, defers or escalates the action.
8. Specify monitoring, false-alarm review and fallback procedures.

Verification

Complete every acceptance item in evidence-checklist.md. Submit the worksheet plus calculations, exported tool output or screenshots named in the worksheet.

Safety, privacy and responsible AI

Use only the supplied scenario data unless your organisation has authorised another dataset. Remove personal, confidential and export-controlled information. AI outputs are suggestions until verified by the named human decision owner.